



Software Release Notes for the Affirmed Private Network Service

Release 1.3.0.0

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Release Abstract

This document describes the new features/enhancements, restrictions, and anomalies for the Affirmed Networks Affirmed Private Network Service, Release 1.3.0.0.

Installation and Upgrade Path

Installation and upgrade instructions can be found in the *Affirmed Private Network Service Deployment Guide*.

To install APNS, your Azure subscription must be enabled by both Azure Network Function Manager Service and NetFoundry. Please [Email support](#) for assistance with this process.

Software Components

Release 1.3.0.0 consists of the following software components:

- PNSM version 1.4
- EMS version 12.2.1
- ASAP vTransactor version 7.3
- MCC version 13.0.0.5-2008
- MME version 15.0.20.15
- HPE HSS PNW-LTE version 04.03.02
- Azure Network Function Manager service
- Azure Stack Edge versions 2110 (**2.2.1758.4034**) , 2111 (**2.2.1777.4088**) and 2203 (**2.2.1902.4561**)

New Features and Enhancements

This release contains security enhancements for the APNS software application.

Constraints

The following constraints apply to this release.

Constraint	Description
One Private Network Service Manager Instance must exist in the cloud for each Enterprise Customer.	

VPN connectivity is required between the Private Network Service Manager and Azure Stack Edge on the customer premises.

NetFoundry's Zero Trust VPN solution can be used to enable the communication between the Service Manager and VMs on ASE (MCC, MME & HSS). For instructions to set up NetFoundry, please refer to the *Affirmed Private Network Service Deployment Guide*.

Known Anomalies

This section lists the known anomalies in this release.

Anomaly	Work Around
If MCC is deployed as the last VNF in the network creation workflow, interface communication between MCC and MME may not work.	Perform the following steps as a workaround: 1. Stop the MCC and HSS VM from the Azure portal Virtual Machine view. 2. After both VMs are stopped, start the MCC VM. 3. When the MCC VM has started, start the HSS VM.
After HSS VNF upgrade, Configuration backup and restore is not working.	You may need to restart mqtt service and the HSS. To do this enter the following commands: <pre>sudo systemctl stop pnw-lte-mqtt sudo systemctl start pnw-lte-mqtt sudo reboot</pre>
MME Config Backup/Restore should not be triggered when mrm1 is in Active state.	Perform mrm switch over to make mrm0 Active before performing the Config Backup/Restore.
IMSI Import must be done in a batch that does not exceed 10,000.	No more than 10,000 IMSIs should be included during one import file.
Remote File Servers (SFTP) must be used for Network Backup / Restore.	Configure the Remote FTP server in the Global Menu --> Configuration --> File Server --> Settings ,
If there are more than 20k SIMs in the Private Network Service Manager, it is not recommended to perform an HSS audit.	Ensure that SIM Provisioning (activation or deactivation) is not in progress while performing a manual HSS synchronization.
Do not restart the PNSM server when a Distributed Edge or All-In-Cloud deployment is in progress. This causes the deployment to fail.	As a work around, clean up the created sources on the Cloud manually, then re-trigger the deployment after the server restarts.
The Network Upgrade notification may appear in the User Interface despite all Networks existing on the latest packet core version.	Refresh the browser to correct this issue.
Network Elements may still be visible after a Distributed Edge or All-In-Cloud network has been deleted from the Service Manager application.	Refresh the browser to correct this issue.

After creating a Private Network, the following alarms display in the Service Manager User Interface:	You can safely ignore these alarms.
▪ Inter-node connectivity health check over data fabric path B is down. ▪ Inter-node connectivity health check over data fabric path A is down. ▪ Communication over the data fabric is down. ▪ Approaching or crossed licensed service capacity enforcement level. ▪ The cluster needs to be rebooted because one or more of the following have changed on one or more nodes in the cluster. ▪ A request for MCM switchover has failed.	
The Upgrade Notification icon does not display when you log in to the Service Manager Application or after a browser refresh.	Use the global right-click menu option to upgrade the network.
In a fresh network deployment of APNS version 1.2, the ping command does not work from the MCC CLI.	Run the following commands from the MCC CLI: 1. Type ssh-keygen, and press enter for all questions as prompted. 2. Change the directory by entering cd /home/root/.ssh. 3. Enter cat id_rsa.pub >> /home/internal/.ssh/authorized_keys.
MCC rollback may fail from 12.2.1.2-23 to 12.0.0.0-1193.	Run the image-management rollback force command from the MCC CLI.
A direct MME upgrade from 15.0.20.9 to 15.20.15 or direct VNF upgrades from APNS 1.2 to APNS 1.3 will fail.	Please refer to the MME Upgrade Workaround section for the steps you are required to perform prior to upgrading VNFs from APNS 1.2 to APNS 1.3.

MME Upgrade Workaround

This section describes the steps you must complete prior to upgrading MME 15.0.20.9 (APNS 1.2) to MME 15.0.20.15 (APNS 1.3). These steps must be performed in all MME nodes managed by the Service Manager software before you attempt any VNF upgrades from APNS 1.2 to APNS 1.3.

1. Download the MME 15.0.20.12 (epc-15.0.20.12.tar) package from the customer's FTP site to the /usr/local/epc/software folder on all MME nodes.
2. Create a tmp folder in all the VMs --> /home/mtc/tmp using the following command: **cd /home/mtc; mkdir tmp**.
3. Copy the epc-15.0.20.12.tar from /usr/local/epc/software folder from active mgmt to all the tmp folders.
4. Execute **tar xvf epc-15.0.20.12.tar -C /home/mtc/tmp --wildcards "libauparse*** in all the tmp folders.

5. Execute `tar xvf epc-15.0.20.12.tar -C /home/mtc/tmp --wildcards "libltdl*" in all the tmp folders.`
6. Execute `tar xvf epc-15.0.20.12.tar -C /home/mtc/tmp --wildcards "libprelude*" in all the tmp folders.`
This command opens/extracts the `libauparse0_1%3a2.8.5-2ubuntu6_amd64.deb`, `libprelude28_5.1.1-5_amd64.deb` and `libltdl7_2.4.6-14_amd64.deb`.
7. Execute `sudo dpkg --force-overwrite --force-depends --force-confold -i libauparse0_1%3a2.8.5-2ubuntu6_amd64.deb.`
8. Execute `sudo dpkg --force-overwrite --force-depends --force-confold -i libltdl7_2.4.6-14_amd64.deb.`
9. Execute `sudo dpkg --force-overwrite --force-depends --force-confold -i libprelude28_5.1.1-5_amd64.deb.`
10. Delete the tmp folder from all VMs.

After you have completed these steps in the MME nodes, you can proceed with the in-service upgrade from the Service Manager user interface.

***Note:** This workaround is required only for the VNF upgrade from APNS 1.2 to APNS 1.3, or MME 15.0.20.9 to 15.0.20.15.*

Problems Fixed

This release only contains security updates for APNS components.